TRIBHUVAN UNIVERSITY

Faculty of Humanities and Social Science

Internship Report on

Mobile-Based Business Inventory Management Application

Submitted To

Department of Humanities and Social Science

Arunima College

Boudha, Kathmandu, Nepal

Under The Supervision of

Mr. ALISH SHRESTHA

In Partial Fulfillment of the requirements for the Bachelor of Computer Applications

Submitted By

Rashik Moktan

Tu Reg: 6-2-142-19-2020

February 2026

CERTIFICATE OF COMPLETION

Walkers Hive

Date: 22 Dec, 2025

CERTIFICATE OF COMPLETION

To Whom It May Concern,

This is to certify that Mr. Rashik Moktan has successfully completed an internship at Walkers Hive Pvt. Ltd. from 24 August 2025 to 10 December 2025.

During the internship period, Mr. Moktan actively participated in assigned responsibilities and demonstrated a strong commitment to learning and professional development. He showed the ability to apply theoretical knowledge into practical tasks, maintained a positive work attitude, and worked collaboratively with team members.

Throughout the internship, Mr. Moktan displayed discipline, responsibility, and a willingness to take on new challenges. His contributions were valuable to the organization, and his performance met our expectations.

We wish him every success in future academic and professional endeavors.

Sincerely,

Walkers Hive

Suman Dhital
Chief Technology Officer (CTO)

MENTOR'S RECOMMENDATION FROM COMPANY

Walkers Hive

Date: 22 Dec, 2025

To Whom It May Concern

Subject: Mentor's Recommendation Letter for Mr. Rashik Moktan

I am writing to highly recommend Mr. Rashik Moktan for future academic and professional opportunities. During his internship, I had the opportunity to serve as his mentor and work closely with him, allowing me to observe his professional growth, technical abilities, and work ethic firsthand.

Throughout his internship period, Mr. Rashik Moktan consistently demonstrated strong dedication, responsibility, and a willingness to learn. He approached assigned tasks with enthusiasm and showed the ability to adapt quickly to new challenges. His commitment to meeting deadlines and maintaining quality standards made him a reliable and valuable member of our team.

In addition to his technical competence, Mr. Rashik Moktan possesses excellent communication and collaboration skills. He interacted effectively with team members and stakeholders, contributing positively to a professional and productive working environment.

I strongly recommend him without reservation for any academic program, internship, or professional opportunity he chooses to pursue. Should you require any further information regarding his performance or qualifications, please feel free to contact me.

Sincerely,

....

Suman Dhital

Chief Technology Officer (CTO)

Walkers Hive

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TRIBHUVAN UNIVERSITY

Faculty of Humanities and Social Sciences

Arunima College, Boudha, Kathmandu

SUPERVISOR'S RECOMMENDATION

I hereby recommend that this internship report prepared under my mentorship by **Mr. Rashik Moktan** entitled "**Mobile-Based Business Inventory Management Application**" in the partial fulfilment of the requirements for the degree of BCA in Computer Application of Tribhuvan University and be processed for the evaluation.

As his supervisor during the front-end developer internship program at **Walkers Hive Pvt. Ltd.,** I had the pleasure of working closely with him and observing his growth and development as a front-end developer. During the internship, he consistently demonstrated a strong aptitude for the CSS framework and web development in general. He was able to effectively utilize the various features of Tailwind CSS and React Native to design and build Mobile applications, and he consistently sought to expand his knowledge and skills through independent learning and project work.

……………………….

Mr. Alish Shrestha

Supervisor

Arunima College

Boudha, Kathmandu

TRIBHUVAN UNIVERSITY

Faculty of Humanities and Social Science

Arunima College

LETTER OF APPROVAL

This is to certify that this internship report prepared by **Rashik Moktan** entitled **“Mobile-Based Business Inventory Management Application”**, submitted in partial fulfilment of the requirements for the degree of Bachelor of Computer Application (BCA) of Tribhuvan University, has been evaluated. In our opinion, it is satisfactory in scope and quality as an internship work for the required degree.

…………………………. **Project Supervisor** Mr. Alish Shrestha Arunima College, Boudha	………………………………. **Coordinator** Mr. Bipul Raj Manandhar Arunima College, Boudha
…………………………. **Internship Mentor**	………………………. **External Examiner**

ACKNOWLEDGEMENTS

I would like to express my sincere gratitude to all those who have contributed to the successful completion of my internship program.

First and foremost, I am deeply thankful to my intern supervisor, **Mr. Suman Dhital**, at **Walkers Hive Pvt. Ltd.**, for giving me the valuable opportunity to intern at the organization. I am truly grateful to all the mentors and staff members for their constant guidance, cooperation, direction, and unwavering support throughout the entire internship period.

I extend my heartfelt thanks to our respected project supervisor and esteemed lecturer, **Mr. Alish Shrestha**, Department of Humanities and Social Science, for his insightful suggestions, continuous encouragement, motivation, and expert guidance that played a vital role in completing this report successfully.

I would also like to express my sincere appreciation to **Mr. Bipul Raj Manandhar**, Head of Department, Department of Humanities and Social Science, for his generous support, valuable time, and constructive feedback, which greatly contributed to the quality of this work.

The accomplishment of this internship report would not have been possible without the collective efforts and assistance from many individuals. I am fortunate to have received ongoing support from the entire faculty, and I sincerely acknowledge their contributions.

Finally, I wish to convey my deepest gratitude to my friends, family, and everyone else who supported me directly or indirectly during this internship journey.

Rashik Moktan
6-2-142-19-2020

ABSTRACT

The project focuses on the structured development of a Mobile-Based Business Inventory Management Application powered by the Laravel framework and MySQL database. The primary goal is to design a secure, scalable, and efficient mobile solution that effectively tracks customers, vendors, item flows, stock levels, sales records, and business transactions to meet real-world inventory management needs.

The development process includes project planning, requirement analysis, robust database design using phpMyAdmin, backend implementation with Laravel's MVC architecture, authentication, and RESTful APIs, along with a React Native frontend for cross-platform compatibility and real-time synchronization. Performance optimization, security practices, thorough testing, and debugging ensure a reliable, user-centered application that streamlines business operations.

Keywords: Inventory Management, Mobile Application, React Native, RESTful API

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LIST OF ABBREVIATIONS

API	Application Programming Interface
CRUD	Create, Read, Update, Delete
JSON	JavaScript Object Notation
PHP	Hypertext Preprocessor
REST / RESTful	Representational State Transfer
UI	User Interface
UX	User Experience

CHAPTER 1: INTRODUCTION

1.1 Introduction

In the modern digital era, businesses are rapidly adopting digital technologies to improve efficiency (Laudon K. C., 2022) and manage operations more effectively. One important aspect of business management is inventory and transaction tracking, which helps organizations monitor stock levels, sales, purchases, and overall business activities (Donald J. Bowersox, 2013). However, many small and medium enterprises (SMEs) still rely on manual record-keeping methods or outdated systems (OECD, Paris), which often lead to inefficiencies, errors, and difficulty in accessing real-time information. As businesses grow, these limitations make it challenging to maintain accurate records and make informed decisions (Reynolds, Boston).

This internship was conducted at Walkers Hive Pvt. Ltd., an IT solutions company based in Kathmandu, Nepal, which specializes in web and mobile application development, UI/UX design, startup MVP development, custom software solutions, enterprise systems, and digital services. The company focuses on building scalable and practical digital products that help startups and businesses streamline their operations. Through its expertise in modern technologies, the organization provides innovative software solutions tailored to the needs of entrepreneurs and growing companies.

During the internship, I participated in the development of a Mobile-Based Business Inventory Management Application designed to simplify the management of customers, vendors, products, stock levels, and business transactions. The system was developed as a cross-platform mobile solution using the Laravel framework for the backend (Stauffer, Sebastopol, California), React Native for the mobile frontend (Boduch, Birmingham), and MongoDB as the NoSQL database (Shannon Bradshaw, Sebastopol, California). The application allows businesses to track inventory, record sales and purchases, and manage business data in real time through a mobile interface.

1.2 Problem Statement

Many small and medium businesses in Nepal, especially retail shops, wholesalers, and trading firms, still rely on manual ledgers, spreadsheets, or basic desktop software for inventory management. These traditional methods often lead to inaccurate stock tracking, resulting in overstocking or stock shortages. Manual entry of sales, purchases, and transactions is time-consuming and prone to human error. Business owners also lack real-time visibility into inventory and sales data, making decision-making difficult. Managing customer and vendor information and generating reports can also be inefficient. In addition, there are risks of data loss and security issues with poorly managed records. Therefore, there is a need for an affordable, mobile-accessible inventory management system that is secure, scalable, and easy to use. A modern mobile application can help businesses track inventory, manage transactions, and access data in real time.

1.3 Objectives

General Objective

To design, develop, and implement a secure and efficient mobile-based business inventory management application using Laravel, React Native, and MongoDB to streamline inventory tracking, sales, and transaction management for small and medium enterprises.

Specific Objectives

- To analyze business requirements and design a flexible database schema using MongoDB for storing inventory items, customers, vendors, and transactions.
- To implement a robust backend using Laravel's MVC architecture, including RESTful APIs for data communication and secure authentication.
- To develop a cross-platform mobile frontend with React Native, ensuring intuitive UI/UX for real-time data synchronization.
- To incorporate essential features such as stock monitoring, sales/purchase recording, customer/vendor management, and basic reporting.

1.4 Scope and Limitations

The scope and limitations of a project entitled "Internship Management System" are given below:

1.4.1 Scope

The internship focused on the full development lifecycle of the mobile inventory management app, including:

- Requirement gathering and planning.
- Database design and backend API development (Laravel + MongoDB).
- Frontend mobile app development (React Native).
- Integration, authentication, real-time sync, and basic testing. The application supports core modules: dashboard, items/stock, customers/vendors, sales/purchases, and basic transaction history. It targets small/medium business owners and staff using smartphones.

1.4.2 Limitations

- The application is mobile-only (no web/desktop version developed during internship).
- Advanced features like barcode scanning, detailed analytics, multi-warehouse support, or payment gateway integration were not included due to time constraints.
- Testing was limited to emulators and a few real devices; large-scale production deployment and load testing were outside scope.
- MongoDB was used for flexibility, but no sharding/replication setup was implemented.
- The project assumes internet connectivity for full functionality (offline support may be partial or not implemented).

1.5 Report Organization

The report is divided into chapters, each focusing on various aspects of the internship project entitled “Mobile-Based Business Inventory Management Application”.

- **Chapter 1: Introduction**
  Introduces internship, organization, project, problem statement, objectives, scope and limitations, and report structure.
- **Chapter 2: Introduction to Organization**
  Describes Walkers Hive Pvt. Ltd. It’s background, mission, services, organizational hierarchy, working domains, and the internship department/unit.
- **Chapter 3: Background Study and Literature Review**
  Covers key technologies (Laravel, React Native, MongoDB) and reviews related inventory management systems and existing solutions.
- **Chapter 4: Internship Activities**
  Details of the intern’s roles and responsibilities, weekly activity log, project description, tasks performed, tools, technologies, and methodologies used.
- **Chapter 5: Conclusion and Learning Outcomes**
  Summarizes the internship experience, achievements, and learning outcomes in technical skills, professional skills, and personal growth.

CHAPTER 2: INTRODUCTION TO ORGANIZATION

2.1 Organization Details

Figure 2.1: Organization Logo

Walkers Hive is a Nepal-based technology company dedicated to building strong digital partnerships that evolve with clients' visions. The company has been empowering startups, entrepreneurs, and growing businesses to transform bold ideas into impactful digital products.

Walkers Hive's mission is to create innovative and high-quality software solutions that empower businesses and individuals, enhancing their productivity, efficiency, and overall success. The company strives to deliver cutting-edge technology and exceptional user experiences through our talented team of software developers, engineers, and designers.

The company's vision is to be a globally recognized leader in software development, renowned for our technical excellence, innovation, and customer-centric approach. We strive to revolutionize industries and empower organizations of all sizes through transformative software solutions.

2.2 Organizational Hierarchy

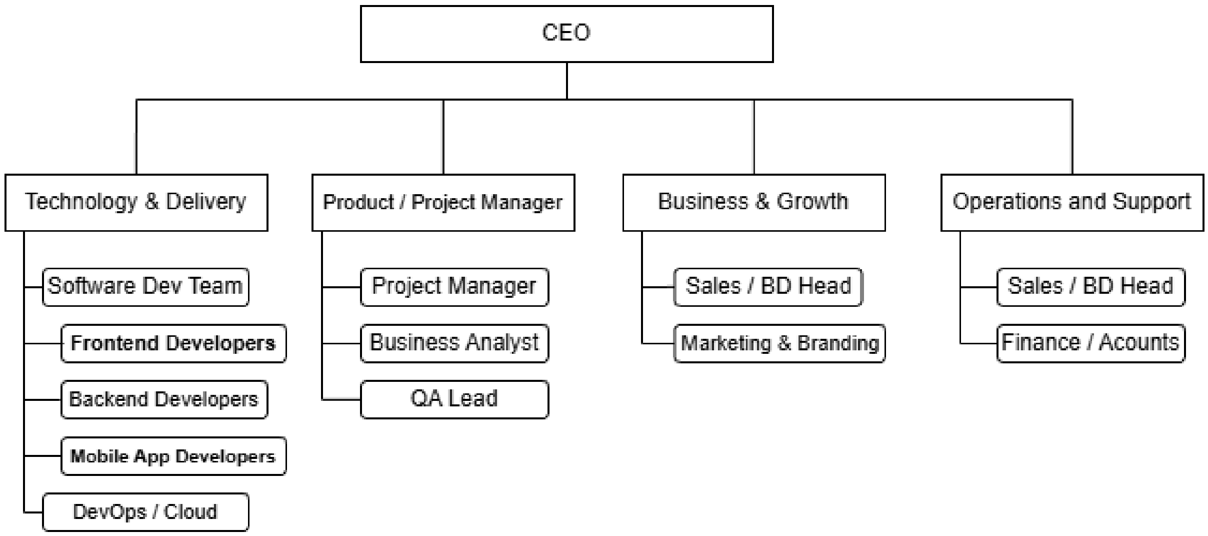

Figure 2.2: Organization Hierarchy

2.3 Working Domain of Organization

Walkers Hive Pvt. Ltd. operates across two complementary domains—technical and business-oriented—allowing the company to provide complete end-to-end digital solutions for startups, small and medium enterprises (SMEs), and growing businesses in Nepal.

- **Web and Mobile App Development:**
  It specializes in building scalable and customizable web and mobile applications tailored to client requirements, including Minimum Viable Product (MVP) and full scale digital solutions.
- **UI//UX Design and Frontend Development:**
  It provides user interface and user experience design services focused on visually engaging and user-centric digital interfaces that enhance usability and accessibility.
- **Custom Software Development:**
  It offers custom software solutions across diverse technologies including PHP Laravel, .NET, Python, Flutter, and Android development to meet specific business needs.
- **Digital Marketing and IT Consulting:**
  Walkers Hive assists businesses with digital strategy and marketing alongside technical consulting to support growth and online presence.

• Project and Product Strategy:

The company supports project planning, technical architecture, and iterative development in collaboration with clients, helping guide technology choices and implementation strategies.

• CMS and Backend System Integration:

The organization builds content management systems and backend structures that allow clients to manage data and content efficiently with secure authentication and data handling modules.

2.4 Description of Intern Department/Unit

The internship was carried out in the Software Development Unit (primarily the Mobile/Web Development Team), which falls under the Technology & Delivery pillar of Walkers Hive Pvt. Ltd.'s organizational structure.

This unit is responsible for the core execution of client projects and internal products, including:

- Full-stack and mobile application development.
- Backend API design and implementation (e.g., using Laravel/PHP).
- Frontend and cross-platform mobile development (e.g., React Native, Flutter).
- Database integration (both relational and NoSQL like MongoDB).
- Collaboration on MVP builds, custom software, and SaaS solutions.

The team operates in a collaborative, agile-inspired environment with direct oversight from the CTO/Tech Lead (Mr. Suman Dhital served in this supervisory/mentorship role during my internship). Interns are integrated closely with senior developers, project managers, and QA members, allowing hands-on participation in real client-facing or product development work.

Key characteristics of the unit:

- Small, dynamic team structure (aligned with the company's 11–50 employee size).
- Emphasis on dedicated developer assignments per project for focused delivery.
- Regular code reviews, pair programming, and mentor guidance to support learning.

During my internship, I was assigned to this unit to work specifically on the Mobile-Based Business Inventory Management Application. This placement provided excellent exposure to end-to-end development processes—from requirement analysis and backend setup to frontend integration and testing—within a supportive, startup-like culture that encourages interns to contribute meaningfully while learning industry practices.

Table 2.1: Internship Details

Particular	**Details**
Start Date of Internship	24th August 2025
End Date of Internship	10th December 2025
Total Duration	3 months
Office Hours	11:00 AM – 6:00 PM
Daily Working Hours	8 hours
Working Days per Week	6 days
Weekly Holiday	Saturday
Position / Role	Frontend Developer

CHAPTER 3: BACKGROUND STUDY AND LITERATURE REVIEW

3.1 Background Study

The developed Mobile-Based Business Inventory Management Application relies on a modern full-stack technology combination that balances performance, scalability, developer productivity, and cross-platform compatibility.

- **Laravel Framework**

  It is a popular open-source PHP web application framework following the Model-View-Controller (MVC) architectural pattern (LLC, 2024). It provides elegant syntax, built-in features for routing, authentication, ORM (Eloquent), middleware, RESTful API development, and security mechanisms (e.g., Sanctum or Passport for token-based auth). In this project, Laravel served as the backend, handling business logic, API endpoints for CRUD operations on inventory data, user authentication, and secure data processing.

- **React Native**

  It is an open-source framework developed by Meta (formerly Facebook) for building native mobile applications using JavaScript and React (Platforms, 2024). It enables a single codebase to target both Android and iOS platforms, with near-native performance through direct use of native components. Key libraries used in this project likely included React Navigation (for screen routing), Axios (for API calls), and possibly state management tools (e.g., Redux or Context API). React Native powered the mobile frontend, delivering an intuitive UI for stock tracking, sales entry, customer/vendor management, and real-time updates.

- **MongoDB (NoSQL Database)**

  It is a document-oriented NoSQL database that stores data in flexible, JSON-like BSON documents (Inc., 2024). Unlike traditional relational databases (SQL), it offers schema flexibility—ideal for inventory systems where items may have varying attributes (e.g., size, color, variants, custom fields). It supports rich querying, indexing, aggregation pipelines, and horizontal scaling.

- **RESTful APIs**

  It is the communication between frontend and backend follows REST principles (Richardson & Amundsen, 2013): stateless, resource-based endpoints (e.g., GET /api/items, POST /api/sales), standard HTTP methods (GET, POST, PUT, DELETE), JSON data format, and proper status codes. This architecture ensures loose coupling, scalability, and ease of future extensions (e.g., adding a web client).

3.2 Literature Review

1. **Current State of Customer–Vendor Management Systems**

   Digital transformation is reshaping customer–vendor management (Laudon & Laudon, anagement Information Systems: Managing the Digital Firm). Research indicates that mobile apps that simplify business operations and provide easy access to information significantly reduce administrative workload and improve communication efficiency compared to traditional method. Many existing systems are overly complex or lack full mobile support, emphasizing the need for simplified, mobile-first solutions.

2. **Technology Trends in Business Management Applications**

   User experience (UX) has a critical influence on mobile application adoption and engagement. A systematic review shows that intuitive navigation, responsive layouts, and performance enhancements are key design elements that affect mobile UX positively across varied application scenarios (Discover Applied Sciences literature review on MAUX, 2025). Prioritizing these features improves workflow efficiency and user satisfaction.

3. **Impact of Digital Solutions on Customer-vendor Management**

   Mobile applications that support interactive and dynamic interfaces help users manage customer and vendor interactions more effectively than manual, spreadsheet-based methods (Chaffey, 2019). Studies in mobile commerce show that quality mobile interfaces contribute to enhanced customer satisfaction and stronger long-term relationships between users and organizations (PMC study on mobile commerce service quality, 2022). Mobile app adoption encourages frequent engagement with vendors and customers, leading to more stable relationships.

4. **Best Practices in Frontend Development for Mobile Applications**

   Mobile UX research highlights that user-centric design practices—such as clear task flows, consistent styling, and interactive feedback components—greatly enhance engagement and usability. For example, best practice guidelines recommend implementing micro-interactions and standardized UI components to increase activation and daily engagement rates (Mobile UX design principles analysis, Spring/Summer 2024). These practices align with component-based architecture in React Native.

5. **Performance Optimization in Mobile Applications**

   Performance is an essential factor in user retention and experience. Industry insights note that optimized performance—such as fast screen load times and smooth transitions—is vital to maintaining engagement and preventing task abandonment in mobile applications. Users are less likely to continue using apps that lag or respond slowly.

6. **User Experience and Accessibility Considerations**

   Usability and accessibility strongly influence widespread adoption of mobile apps. Research on mobile UX indicates that features such as readability, intuitive navigation, and responsive design reduce cognitive load and increase task success rates for users of all abilities. Prioritizing accessibility improvements leads to higher adoption and satisfaction.

7. **Mobile-First Approaches in Professional Applications**

   Mobile-first design prioritizes simplicity and essential functionality, leading to higher engagement and retention compared to traditional desktop-centric designs. Findings from design strategy studies show that mobile-first approaches result in improved user satisfaction, reduced bounce rates, and enhanced loyalty in app environments. These trends support developing mobile-oriented business tools like Optic Soft Network.

8. **Future Trends in Customer–Vendor Management Applications**

   Emerging research emphasizes real-time features, push notifications, and adaptive interfaces as future drivers of mobile app success. Studies involving notification timing and user behavior models show that strategically designed real-time systems.

CHAPTER 4: INTERNSHIP ACTIVITIES

4.1 Roles and Responsibilities

As Full-Stack Development Intern at Walkers Hive Pvt. Ltd., my primary responsibilities revolved around contributing to the development of the Mobile-Based Business Inventory Management Application under the supervision of Mr. Suman Dhital (intern supervisor and CTO/Tech Lead).

Key roles included:

- Assisting in requirement analysis and feature planning for the inventory system.
- Designing and implementing database schemas using MongoDB.
- Developing backend RESTful APIs with Laravel (MVC architecture, authentication, CRUD operations).
- Building cross-platform mobile UI/UX screens with React Native.
- Integrating frontend with backend via API calls and handling real-time data sync.
- Performing testing, debugging, and applying basic security practices.
- Participating in code reviews, team discussions, sprint planning, and documentation.

4.2 Weekly Log

The internship spanned 8–12 weeks / from Bhadra-20 to Mangsir 25. Below is a summarized week-by-week log of major activities (detailed logs/sheets in Appendices).

Week 1–2

Orientation, requirement gathering, and project planning. Studied existing inventory needs, finalized features (customers, vendors, items, stock, sales/purchases), set up development environment (Laravel, React Native, MongoDB).

Week 3–4

Database design and backend setup. Created MongoDB collections/schemas for entities, implemented Laravel models/controllers/routes for CRUD, added authentication (e.g., Sanctum tokens).

Week 5–6

API development and testing. Built RESTful endpoints (e.g., /api/items, /api/sales), tested with Postman, handled validation/errors.

Week 7–8

Mobile frontend development. Set up React Native project, created screens (dashboard, item list, add sale/purchase), integrated API calls (Axios), implemented navigation and state management.

Week 9–10

Integration and real-time features. Connected frontend-backend sync, added basic user roles (admin/staff), performed device testing (emulators/real phones).

Week 11–[end]

Debugging, optimization, final testing, documentation, and demo preparation. Fixed bugs, improved UI responsiveness, prepared report materials.

4.3 Description of the Projects Involved During Internship

The main project was the Mobile-Based Business Inventory Management Application—a cross-platform mobile app for SMEs to track inventory, sales, purchases, customers, vendors, and transactions in real time.

- **Backend**: Laravel (MVC) + MongoDB (via PHP driver/package).
- **Frontend**: React Native (for Android/iOS). s
- **Communication**: RESTful APIs (JSON format).
- **Core Features:** Dashboard overview, item/stock management (add/edit/delete, low-stock alerts), customer/vendor profiles, sales/purchase recording, transaction history, basic search/filter, user login/roles.
- **Tools:** VS Code, Git (for version control), Postman, MongoDB Compass, Expo (for React Native dev/testing), Android Studio/iOS simulator.

4.4 Tasks / Activities Performed

During my internship, I was assigned to work on various tasks and activities related to frontend as well as backend API Integration.

Major tasks included:

- Requirement analysis and wireframing (basic sketches for mobile screens).
- MongoDB schema design (collections for Items, StockTransactions, Customers, Vendors, Sales, Purchases).
- Laravel backend: Models, Controllers, Routes, Middleware for auth/validation, Eloquent-like queries via MongoDB package.
- React Native frontend: Components (Item Card, SaleForm), screens with React Navigation, API integration (fetch), state handling.
- Security: Token-based auth, input sanitization, HTTPS considerations.
- Testing: Unit/integration tests (basic), manual testing on devices, bug fixing.
- Other: Git commits/pushes, code reviews with mentor, documentation (API docs, README).
- Methodologies: Agile inspired (weekly sprints, daily stand-ups via chat), version control with Git, collaborative tools (e.g., WhatsApp/Google Meet for reviews, as per company process).

CHAPTER 5: CONCLUSION AND LEARNING OUTCOMES

5.1 Conclusion

The internship at Walkers Hive Pvt. Ltd. was a highly rewarding and transformative experience that successfully bridged academic knowledge with real-world software development practices. Over the course of the internship, I was able to contribute meaningfully to the development of a Mobile-Based Business Inventory Management Application using Laravel (backend), React Native (frontend), and MongoDB (database). The application achieved its core objectives: providing a secure, cross-platform mobile tool for small and medium enterprises to track inventory, customers, vendors, sales, purchases, and transactions in real time.

Key achievements include:

- Completion of a functional full-stack mobile application with RESTful API integration and real-time data synchronization.
- Implementation of flexible MongoDB schemas suitable for varying product attributes common in Nepali retail/wholesale businesses.
- Delivery of an intuitive React Native interface tested on both Android and iOS emulators/devices.
- Adherence to industry best practices such as secure authentication, input validation, version control with Git, and iterative debugging.

The project not only addressed the identified problem of manual and inefficient inventory management in SMEs but also demonstrated the practical value of modern technologies in creating affordable, mobile-first solutions. The internship reinforced Walkers Hive’s commitment to empowering startups and local businesses through innovative, client-focused digital products. Overall, the experience met and exceeded expectations, confirming the effectiveness of collaborative mentorship in a professional IT environment.

5.2 Learning Outcomes

Technical Skills

- Gained hands-on proficiency in full-stack mobile application development using Laravel (MVC, RESTful APIs, authentication), React Native (components, navigation, state management, API integration), and MongoDB (document modeling, queries, schema design).
- Learned practical integration of NoSQL databases with PHP frameworks and handling JSON/BSON data flows.
- Developed debugging, testing (Postman, device emulators), and optimization skills for mobile performance and API reliability.
- Became familiar with Git-based version control, code reviews, and collaborative development workflows.

Professional Skills

- Improved time management and task prioritization through weekly milestones and iterative development cycles.
- Enhanced communication skills via regular discussions with mentors (Mr. Suman Dhital), code reviews, and progress reporting.
- Understood agile-like processes in a real IT company setting, including requirement analysis, sprint planning, and client-focused delivery.
- Gained exposure to industry standards for documentation, security practices, and quality assurance.

Personal Growth

- Developed greater confidence in tackling complex technical problems independently while seeking guidance when needed.
- Cultivated adaptability and resilience in a fast-paced startup-like environment with evolving requirements.
- Realized the importance of user-centered design and how small features can significantly impact business usability.

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APPENDIX

A.1 Results

Figure A.1: App Login Screen

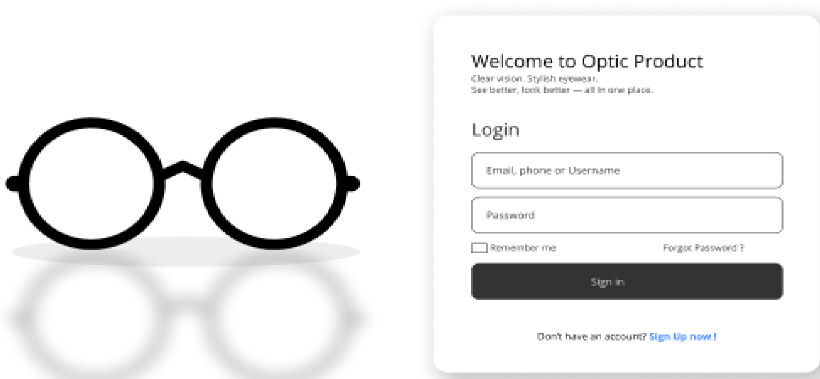

Figure A.2: Web Login page

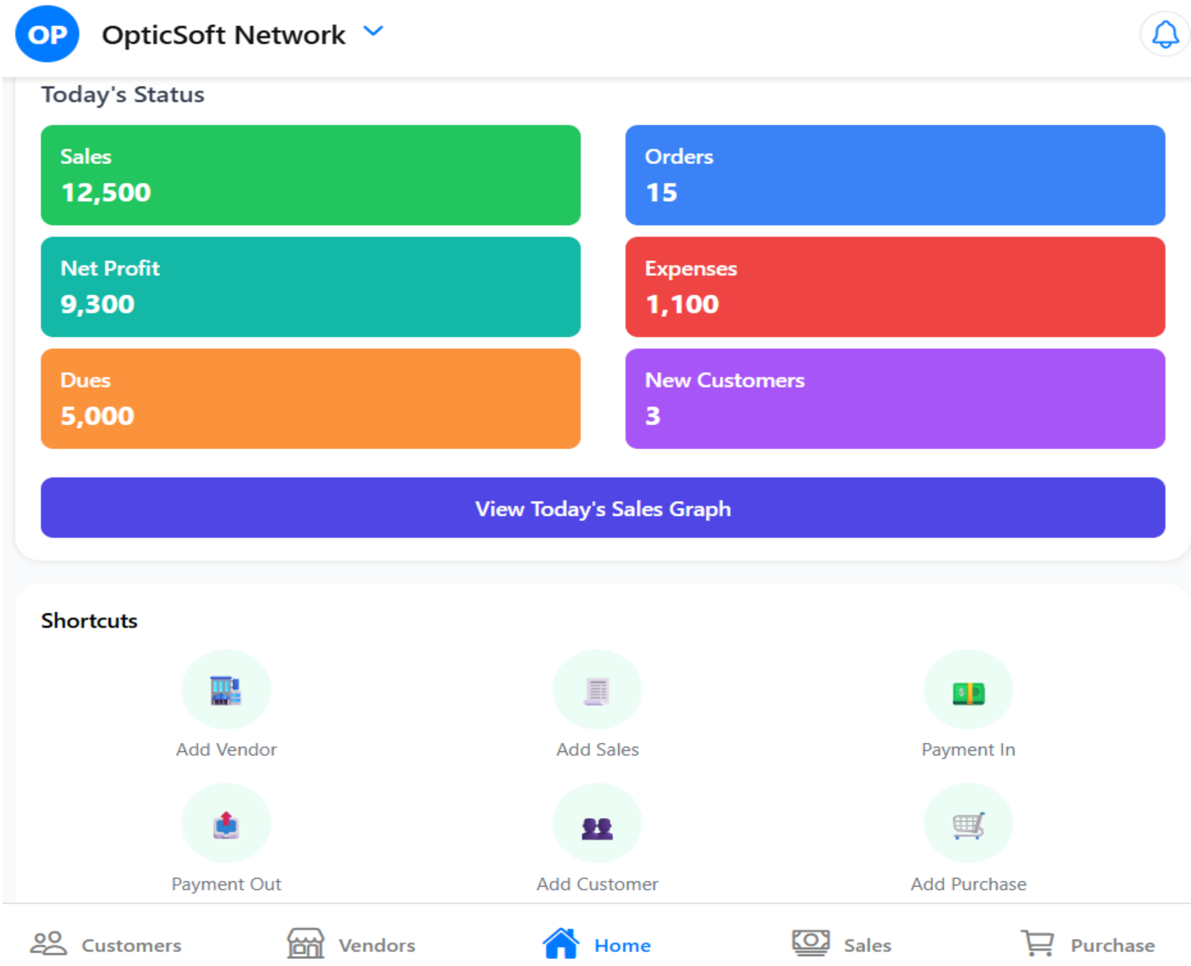

Figure A.3: Mobile Dashboard Screen

A.2 Source Code

```
app
  modules
    auth
      api
      screens
      store
    customer
    dashboard
    notification
    purchase
    sales
    vendor
  navigation
  screens
  shared
```

Figure A.4: App Modules Structure

```
// Sample data
const STATS = [
  { id: "1", label: "Sales", value: "12,500", bgClass: "#22C55E" },
  { id: "2", label: "Orders", value: "15", bgClass: "#3B82F6" },
  { id: "3", label: "Net Profit", value: "9,300", bgClass: "#14B8A6" },
  { id: "4", label: "Expenses", value: "1,100", bgClass: "#EF4444" },
  { id: "5", label: "Dues", value: "5,000", bgClass: "#FB923C" },
  { id: "6", label: "New Customers", value: "3", bgClass: "#A855F7" },
];

const SHORTCUTS = [
  { id: "s1", emoji: "➕", label: "Add Vendor", screen: "AddVendor" },
  { id: "s2", emoji: "🧾", label: "Add Sales", screen: "AddSales" },
  { id: "s3", emoji: "💵", label: "Payment In", screen: "PaymentIn" },
  { id: "s4", emoji: "📤", label: "Payment Out", screen: "PaymentOut" },
  { id: "s5", emoji: "➕", label: "Add Customer", screen: "AddCustomer" },
  { id: "s6", emoji: "🛒", label: "Add Purchase", screen: "AddPurchase" },
];
```

Figure A.5: Preset Data for UI Test

```
return (
  <SafeAreaView style={styles.container}>
    {/* Fixed Header */}
    <Header
      textLogo="OP"
      title="OpticSoft Network"
      toolSymbol="🔔"
      navi="Notification"
      main={true}
    />

    {/* Scrollable Content with pull-to-refresh */}
    <FlatList
      data={CUSTOMERS}
      keyExtractor={(item) => item.id}
      renderItem={({ item }) => <CustomerItem item={item} />}
      ListHeaderComponent={
        <View>
          <StatusSection
            stats={STATS}
            onViewGraph={() => console.log("View Graph clicked")}
          />
          <ShortcutsSection shortcuts={SHORTCUTS} />
        </View>
      }
      contentContainerStyle={styles.listContent}
      showsVerticalScrollIndicator={false}
      refreshControl={
        <RefreshControl
          refreshing={refreshing}
          onRefresh={onRefresh}
          colors={["#3B82F6"]}
          tintColor="#3B82F6"
        />
      }
    />
  </SafeAreaView>
);
```

Figure A.6: Returning Dashboard Widgets Function

```
function TabScreens() {
  const insets = useSafeAreaInsets(); // Detect gesture/nav button insets

  return (
    <Tab.Navigator
      initialRouteName="Home"
      screenOptions={({ route }) => {
        const tab = TABS.find((t) => t.name === route.name);
        return {
          tabBarIcon: ({ focused, color, size }) => {
            const iconName = focused
              ? tab?.icons.active
              : tab?.icons.inactive || "help-circle-outline";
            return <Ionicons name={iconName} size={size} color={color} />;
          },
          tabBarActiveTintColor: "#007AFF",
          tabBarInactiveTintColor: "gray",
          tabBarStyle: {
            height: 60 + insets.bottom, // dynamic height for gesture/button nav
            paddingBottom: insets.bottom > 0 ? insets.bottom : 8,
            paddingTop: 5,
          },
          headerShown: false,
        };
      }}
    >
      {TABS.map((tab) => (
        <Tab.Screen key={tab.name} name={tab.name} component={tab.component} />
      ))}
    </Tab.Navigator>
  );
}
```

Figure A.7: Main Screen bottom navigation

A.3 Log Tables

Internship Attendance Sheet

Organization Name: Walkers Hive Pvt. Ltd

Internship Period: 24^{th} Aug – 10^{th} Dec

Student Name:

Week-1 (August)

Date	Day	Time In	Time Out	Activities Performed
24th	Sun	11AM	6PM	Company orientation, introduction to development team, overview of internship
25	Mon	11Am	6PM	Installed development tools, React Native environment, Android studio & Git.
26	Tue	11AM	6PM	Introduction to React Native fundamentals.
27	wed	11AM	6PM	learned basic UI component & JSX syntax while building small project.
28	Thu	11AM	6PM	Introduction to figma & understanding the UI design for the shop management.
29	Fri	11AM	6PM	Cloned the mobile app project from Github and successfully ran it. on
30	Sat	Day off		

Student Signature :

Mentor Signature:

Week-2 (Aug-Sep)

Date	Day	Time In	Time Out	Activities Performed
31	Sun	11AM	6PM	Studied the existing project codebase & component structure.
1 Sep	Mon	11AM	6PM	Practiced creating reusable UI components used across the app.
2 Sep	Tue	11AM	6PM	Implemented basic screen layout using flexbox styling.
3 Sep	wed	11AM	6PM	learned Navigation between screens using React Navigation.
4 Sep	Thu	11AM	6PM	Started working on project item listing screen UI
5 Sep	friday	11AM	6PM	Tested UI components & fixed layout.
6 Sep	Sat	Day off		

Student Signature :

Mentor Signature :

Week-3 (September)* protest week.

Date	Day	Time In	Time Out	Activities Performed
7	Sun	11 AM	6 PM	Implemented item list display using Flatlist
8	Mon	11 AM	6 PM	learned how to manage component state using useState hook
9	Tue	11 AM	6 PM	Added input fields for adding new shop items.
10	Wed	11 AM	6 PM	Implemented basic form validation for item entry.
11	Thu	11 AM	6 PM	worked on improving UI layout according to figma design.
12	Fri	11 AM	6 PM	fixed minor UI bugs and committed changes to git
13	Sat			Day off

Student Signature :

Mentor Signature:

Week-4 (September)

Date	Day	Time In	Time Out	Activities Performed
14	Sun	11 AM	6 AM	introduction to API Integration for retrieving data.
15	Mon	11 AM	6 PM	Fetched item data from backend API & displayed in mobile app.
16	Tue	11 AM	6 PM	Implemented loading indicators while fetching data.
17	Wed	11 AM	6 PM	Worked on error handling for API
18	Thu	11 AM	6 PM	Improved item list UI & added item detailed view.
19	Fri	11 AM	6 PM	Code cleanup & git commits
20	Sat			Day off

Student Signature :

Mentor Signature:

Week-7 (october)

Date	Day	Time In	Time Out	Activities Performed
5	Sun	11AM	6PM	Started working on purchase Module.
6	Mon	11AM	6PM	Designed purchase list screen based on Figma design.
7	tue	11AM	6PM	Implemented purchase entry from UI
8	wed	11AM	6PM	Integrated total purchase recording functionality
9	thu	11AM	6PM	Test purchase data with backend API
10	fri	11AM	6PM	Debugged errors and improved UI flow.
11	Sat	Day off		

Student Signature :

Mentor Signature:

Week-8 October

Date	Day	Time In	Time Out	Activities Performed
12	Sun	11AM	6PM	Implemented customer list management feature.
13	Mon	11AM	6PM	Created customer registration form.
14	tue	11AM	6PM	Connected customer data with backend service.
15	wed	11AM	6PM	Displayed customer list using Flatlist
16	thu	11AM	6PM	Improved input validation and UI design.
17	fri	11AM	6PM	Fixed minor bugs in customer module.
18	Sat	Day off		

Student Signature :

Mentor Signature:

Week-11 (November)

Date	Day	Time In	Time Out	Activities Performed
2	Sun	11AM	6PM	worked on dashboard Screen
3	Mon	11 AM	6 PM	Displayed summary data such as item count & vendor records
4	tue	11 AM	6PM	Implemented API calls for dashboard statistics
5	wed	11 AM	6PM	improved dashboard UI layout
6	thu	11AM	6 PM	Tested dashboard functionality
7	Fri	11 AM	6PM	fixed UI alignment issues.
8	Sat	Day off		

Student Signature :

Mentor Signature:

Week-12 (November)

Date	Day	Time In	Time Out	Activities Performed
9	Sun	11 AM	6PM	Implemented edit & update functionality for items.
10	Mon	11AM	6PM	added edit feature for vendor information
11	tue	11 AM	6PM	Tested update functionality across modules
12	wed	11 AM	6PM	Fixed bugs related to data updates
13	thu	11AM	6PM	improved UI feedback message
14	friday	11AM	6PM	Code review with supervisor
15	Sat	Day off		

Student Signature

Mentor Signature:

Week-13 (November)

Date	Day	Time In	Time Out	Activities Performed
16	Sun	11AM	6PM	learned techniques for improving React Native performance
17	Mon	11AM	6PM	optimized component rendering & state updates
18	tue	11AM	6PM	Improved code structure & Modularization.
19	wed	11AM	6PM	Applied mentor feedback on code improvement.
20	thu	11AM	6PM	Tested app performance on different device.
21	Fri	11AM	6PM	update project document.
22	Sat	Day off		

Student Signature :

Mentor Signature:

Week-14 November

Date	Day	Time In	Time Out	Activities Performed
23	Sun	11AM	6PM	worked on overall app navigation.
24	Mon	11AM	6PM	fixed layout responsive issues.
25	tue	11AM	6PM	improved consistency across UI screens.
26	wed	11AM	6PM	Conducted functional testing across modules.
27	thu	11AM	6PM	fixed identified bugs
28	fri	11AM	6PM	Prepared progress updated.
29	Sat	Day off		

Student Signature :

Mentor Signature:

Week-15 (Nov-Dec)

Date	Day	Time In	Time Out	Activities Performed
30	Sun	4 AM	6 PM	Continued working on vendor & purchased models
1 Dec	Mon	4 AM	6 PM	Assisted in preparing UI component for item & customer screen.
2 Dec	Tue	11 AM	6 PM	Tested different features & identified bugs in item tracking module
3 Dec	Wed	11 AM	6 PM	Fixed minor UI & functionality issue in existing screen.
4 Dec	Thu	11 AM	6 PM	Discussed pending features & improved
5 Dec	Fri	11 AM	6 PM	continued development & testing of incomplete modules.
6 Dec	Sat	Day off		

Student Signature :

Mentor Signature:

Week-16 December

Date	Day	Time In	Time Out	Activities Performed
7	Sun	11 AM	6 PM	worked on debugging and improving application stability.
8	Mon	11 AM	6 PM	Assisted team member in testing vendor transaction
9	Tue	11 AM	6 PM	updated UI components based on mentor feedback.
10	Wed	11 AM	6 PM	Internship concluded while the project was still under development
11	Thu	—	—	—
12	Fri	—	—	—
13	Sat	Day off		

Student Signature

Mentor Signature: